

HPE Networking Assistant — Test Plan

End-to-end acceptance test plan covering the **installation process** and **every tool and visualizer**. Written to be executed on a single Windows PC acting as the test client.

- **Product version under test:** 1.7.0
- **Repository:** <https://github.com/hpe-networking-lab/hpe-networking-assistant>
- **Two test layers:**
 1. **Automated** — the pytest suite (65 tests) validates all tool logic with a mocked Mist API; no token or network required.
 2. **Manual acceptance** — live tests against a real Mist tenant through Claude Desktop, below.

Mark each case **PASS / FAIL** in the result column and capture notes/screenshots for failures.

1. Test environment

Item	Value for this run
Test client	This Windows PC
OS	Windows 10/11
Python	3.10+ (python --version)
Claude Desktop	Latest
Mist tenant	_Use a lab/non-production org for write tests_
Mist token role (read tests)	Observer / read
Mist token role (write tests)	Network Admin / write
Region	Auto-detected

Safety

- **Read-only tools are safe in production.** Run them anywhere.
- **Write-mode tests (Section 9) change device state/config** (rename, reboot, LED, claim/assign). Run them **only against a lab org** and only on devices you may disrupt.

2. Prerequisites checklist

#	Check	Expected	Result
P1	python --version	3.10 or newer	
P2	Claude Desktop installed and opens	Launches	
P3	Mist API token created (My Account → API Token)	Token copied	
P4	(Optional) Git installed for the dev/automated path	git --version works	

3. Automated test suite (developer path)

Run from a clone of the repo. Validates all tools/visualizers offline.

#	Step	Command	Expected	Result
A1	Clone + install	<code>pip install -e ".[dev]"</code>	Installs with no errors	
A2	Run tests	<code>pytest -q</code>	65 passed	
A3	Byte-compile	<code>python -m compileall src server</code>	No errors	
A4	Validate manifest	<code>python -c "import json;json.load(open('manifest.json'))"</code>	No error	

Coverage map (automated):

Area	Test file
Region normalization + detection	<code>test_regions.py</code> , <code>test_discovery.py</code>
Mist client (auth, pagination, writes, search)	<code>test_mist_client.py</code> , <code>test_write_mode.py</code> , <code>test_client_trace.py</code> , <code>test_access_assurance.py</code>
MCP server / dispatch / tool registry	<code>test_server.py</code>
Onboarding & status	<code>test_onboarding.py</code> , <code>test_status.py</code>
Validation framework	<code>test_validation.py</code>
Reports	<code>test_reports.py</code>
NAC visualizer	<code>test_nac_visualizer.py</code>

4. Installation tests (TC-INST)

#	Objective	Steps	Expected	Result
INST-1	Download artifact	From the latest release, download <code>hpe-networking-assistant-<ver>.dxt</code>	File downloads	
INST-2	Verify checksum (optional)	Compare SHA-256 against <code>SHA256SUMS.txt</code>	Matches	
INST-3	Install extension	Claude Desktop → Settings → Extensions → drag in the <code>.dxt</code>	Extension details + config form appear	
INST-4	Config form is token-only	Inspect the form	Only Mist API Token (required) and Enable write operations (off) are shown — no region/org fields	
INST-5	Enter token + enable	Paste token, leave write off, Install/Enable	Extension enables without error	
INST-6	Token stored securely	—	Token saved to OS keychain (not shown again in plain text)	

#	Objective	Steps	Expected	Result
INST-7	Time budget	From clicking install to first successful query	Under 10 minutes	

5. Onboarding tests (TC-ONB)

#	Objective	Prompt / action	Expected	Result
ONB-1	First-run wizard	New chat: "Set me up."	Claude calls <code>start_setup</code> ; auto-detects region, discovers orgs/sites, runs validation, returns READY FOR USE	
ONB-2	Region auto-detected	Observe report	Correct region shown; you never entered it	
ONB-3	No IDs required	Observe	You are never asked for Org ID / Site ID / API endpoint	
ONB-4	Multi-org prompt	(If token sees >1 org)	Claude lists orgs by name and asks which to use; answering by name completes setup	
ONB-5	Persistence	Restart Claude Desktop, ask "what's my status?"	Region/org remembered (no re-probe needed)	

6. Read-only tool tests (TC-RO)

Use natural-language prompts; verify Claude calls the right tool and returns sensible data.

#	Tool	Prompt	Expected	Result
RO-1	<code>get_status</code>	"Am I in read-only or read-write mode?"	Reports read-only, region, org, account	
RO-2	<code>get_organizations</code>	"What organizations can I access?"	Lists org name(s)	
RO-3	<code>get_sites</code>	"List my sites."	Site names + locations	
RO-4	<code>get_access_points</code>	"Show all APs in my organization." (success-criteria query)	AP list with online/offline counts	
RO-5	<code>get_access_points</code> (site)	"Show APs at <site name>."	APs filtered to that site	
RO-6	<code>get_switches</code>	"List my switches."	Switch list with status	
RO-7	<code>get_clients</code>	"How many clients are connected at <site>?"	Client count/list	
RO-8	<code>get_offline_access_points</code>	"Which access points are offline right now?"	Offline AP list (matches portal)	
RO-9	Cross-check	Compare RO-4/RO-8 counts to the Mist portal	Numbers match	

7. Report tests (TC-RPT)

#	Tool	Prompt	Expected	Result
RPT-1	generate_health_report	"Generate a network health report and save it as a file."	Markdown report (totals, offline APs, per-site) saved as .md	
RPT-2	Health accuracy	Open the report	Counts match RO-4/RO-6/RO-8	
RPT-3	Skip clients	"Generate a health report without the client count."	Report notes clients "not collected" (faster)	
RPT-4	generate_inventory_report	"Create an inventory report of all devices."	Markdown table: every AP+switch with model/serial/MAC/site/status/version	
RPT-5	Convert	"Convert that report to PDF."	Claude produces a PDF from the Markdown	

8. Client trace & Access Assurance tests (TC-CT / TC-AA)

> Pick a real client MAC/hostname from RO-7 for these.

#	Tool	Prompt	Expected	Result
CT-1	find_client (mac)	"Find the client with MAC <mac>."	Site, AP, SSID, IP, band, RSSI, last seen	
CT-2	find_client (hostname)	"Where is hostname <name>?"	Matching client(s) located	
CT-3	trace_client	"Trace client <mac> over the last hour."	Event timeline, per-type counts, highlighted failures	
CT-4	No-input guard	"Find a client." (no mac/hostname)	Friendly error asking for mac or hostname	
AA-1	get_nac_clients	"List NAC clients from the last day."	Authenticated NAC clients (user, type, auth type, SSID, VLAN, rule, status) — _empty if Access Assurance not configured_	
AA-2	Filter	"Show only EAP-TLS NAC clients."	Filtered list	
AA-3	troubleshoot_authentication	"Troubleshoot authentication failures in the last 24 hours."	Event timeline, type counts, failure highlights	
AA-4	Focus by client	"Why is <mac> failing authentication?"	Events scoped to that MAC	

> **Note:** AA tests require a tenant with **Access Assurance (NAC)** enabled. If your tenant has no NAC, expect empty results (not an error). Confirm the NAC events endpoint path against your tenant (see Release Notes 1.6.0).

9. NAC dashboard test (TC-NAC)

#	Tool	Prompt	Expected	Result
NAC-1	generate_nac_dashboard	"Build a NAC dashboard and save it as HTML."	Self-contained .html produced	
NAC-2	Open offline	Double-click the .html	Opens in browser; cards + bar charts render with no internet	

#	Tool	Prompt	Expected	Result
NAC-3	Content	Inspect	Cards (clients, events, failures, success rate) + charts for auth types, client types, status, event types, top failing users/rules	
NAC-4	Empty-state	(No-NAC tenant)	Renders with zeros / "No data" rather than erroring	

10. Write-mode tests (TC-WR) — LAB ORG ONLY

> These change device state/config. Use a lab org and a **write-capable token**.

#	Objective	Prompt / action	Expected	Result
WR-1	Enable write mode	Settings → toggle Enable write operations on; restart prompt	Extension restarts	
WR-2	Write tools appear	"What can you do now?" / list tools	rename_device, reboot_device, locate_device, claim_devices, assign_devices_to_site now available	
WR-3	Write-readiness	"Check my setup." (get_status)	If token is read-only → reports it + guidance; if write-capable → confirms write access	
WR-4	Confirmation gate	"Rename AP <mac> to TEST-AP."	Claude asks to confirm; nothing changes until you confirm	
WR-5	Rename	Confirm WR-4	Device renamed (verify in portal); revert afterward	
WR-6	LED locate	"Blink the locate LED on <mac>." → confirm	LED toggles (verify physically/portal)	
WR-7	Reboot (disruptive)	"Reboot <mac>." → confirm	Device restarts (lab device only)	
WR-8	Read-only token blocked	With a read-only token + write on, attempt a write	Mist returns permission error; guidance shown	
WR-9	Disable write	Toggle off; restart	Write tools no longer offered	

11. Validation framework / CLI (TC-VAL) — developer path

#	Objective	Command	Expected	Result
VAL-1	Validate	hpe-mist-validate (with MIST_API_TOKEN set)	Prints per-check results ending in READY FOR USE	
VAL-2	Bad token	Set an invalid token, run again	REQUIRES ATTENTION , authentication check fails clearly	
VAL-3	Setup wizard CLI	hpe-mist-setup	Auto-detects region, discovers orgs/sites, saves config, runs validation	

12. Negative & edge tests (TC-NEG)

#	Objective	Action	Expected	Result
NEG-1	Invalid token	Install with a bad token; "Set me up."	"Could not validate this token against any Mist cloud" + guidance	
NEG-2	No org access	Token with no org privileges	Clear "no organization access" message	
NEG-3	Multi-org ambiguity	Token spanning orgs; ask for APs without naming org	Claude asks which org by name	
NEG-4	Empty results	Query a site with no devices	Returns zero counts gracefully (no crash)	
NEG-5	Write attempt while read-only mode	With write mode off, ask to rename a device	Write tool not available; Claude explains how to enable	

13. Upgrade test (TC-UPG)

#	Objective	Action	Expected	Result
UPG-1	Install newer .dxt	When a new release exists, install over the old one	Replaces previous version; token/config preserved	
UPG-2	New tools present	Ask for a newly added capability	Works without reconfiguration	

14. Success criteria (project acceptance)

#	Criterion	Result
SC-1	Install Claude Desktop, install the extension, enter a token, and run "Show all APs in my organization" with no manual API configuration	
SC-2	Whole install + first query takes under 10 minutes	
SC-3	Customer never needs to know Org ID, Site ID, or API endpoint	
SC-4	Read-only by default; write mode is opt-in and confirmation-gated	
SC-5	Automated suite: 65 passed	

15. Sign-off

Role	Name	Date	Result (Pass/Fail)	Notes
Tester				

Role	Name	Date	Result (Pass/Fail)	Notes
Reviewer				

Defects found: _log each failure with the test ID, expected vs actual, and a screenshot._